Positive Multilinear Forms on Non-Atomic L_p

Automated Math Discovery Smoke Test

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Source and Open Question

This packet concerns Question 3 from:

> Luis A. Garcia, Jose Lucas P. Luiz, Vinicius C. C. Miranda, *Norm attainment for multilinear operators and polynomials on Banach Spaces and Banach lattices*, arXiv:2605.12117.

The source location is `arxiv.tex:1007--1071`, final section "Open Questions", Question 3. The question appears on page 28 of the source paper:

> We showed in Section 3 that all positive bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0,1])$ are norm-attaining while there are non-norm attaining bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0,1])$. From this and our results, we obtain that all positive bilinear operators $\ell_4 \times \ell_4 \to L_1([0,1])$ are weakly sequentially continuous while there are bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0,1])$ that are not weakly sequentially continuous. It would be interesting, if we get examples with the domain being non-atomic Banach lattices. In view of this, we leave the following question:
>
> **Question 3:** Are all positive n-linear forms on $L_{p_1}[0,1] \times \cdots \times L_{p_n}[0,1]$ with $\sum_{i=1}^{n} \frac{1}{p_i} < 1$ weakly sequentially continuous?

The paper defines an n-linear operator $A : X_1 \times \cdots \times X_n \to Y$ to be weakly sequentially continuous when

$$A(x_{1,k}, \ldots, x_{n,k}) \to A(x_1, \ldots, x_n)$$

whenever $x_{j,k} \xrightarrow{w} x_j$ in X_j for every $j = 1, \ldots, n$.

Remark 1 (Exponent convention)**.** *The rendered question writes $L_{p_i}[0,1]$ and assumes $\sum_i 1/p_i < 1$. We read this in the surrounding finite-exponent convention of the paper, namely $1 < p_i < \infty$. The inequality already excludes $p_i = 1$. A human reviewer should confirm whether the authors intended to include the endpoint $p_i = \infty$; the proof below resolves the finite-exponent formulation.*

Claim

Theorem 1. *Let $n \geq 2$ and let $1 < p_1, \ldots, p_n < \infty$ satisfy*

$$\sum_{i=1}^{n} \frac{1}{p_i} < 1.$$

Then there is a positive continuous n-linear form on

$$L_{p_1}[0,1] \times \cdots \times L_{p_n}[0,1]$$

which is not weakly sequentially continuous. In particular, Question 3 has a negative answer under the finite-exponent interpretation.

Proof. Define

$$A(f_1, \ldots, f_n) = \int_0^1 f_1(t) \cdots f_n(t)\, dt.$$

Since $\sum_i 1/p_i < 1$, choose $s \in (1, \infty]$ such that

$$\frac{1}{s} = 1 - \sum_{i=1}^{n} \frac{1}{p_i}.$$

Holder's inequality, with the remaining factor filled by the constant function $1 \in L_s[0,1]$, gives

$$|A(f_1, \ldots, f_n)| \leq \prod_{i=1}^{n} \|f_i\|_{p_i}.$$

Thus A is a continuous n-linear form. It is positive in the real Banach-lattice sense, because $f_i \geq 0$ for all i implies $f_1 \cdots f_n \geq 0$ almost everywhere.

Let (r_k) be the Rademacher functions on $[0,1]$. For every $1 < p < \infty$, one has $r_k \xrightarrow{w} 0$ in $L_p[0,1]$. Indeed, if q is the conjugate exponent and $h \in L_q[0,1]$, approximate h in L_q by a dyadic step function h_m. For all sufficiently large k, $\int_0^1 h_m r_k = 0$, and Holder's inequality controls $\int_0^1 (h - h_m) r_k$.

Now set

$$x_{1,k} = r_k, \qquad x_{2,k} = r_k, \qquad x_{j,k} = 1 \quad (3 \leq j \leq n).$$

Then

$$x_{1,k} \xrightarrow{w} 0, \qquad x_{2,k} \xrightarrow{w} 0, \qquad x_{j,k} \xrightarrow{w} 1 \quad (3 \leq j \leq n),$$

but

$$A(r_k, r_k, 1, \ldots, 1) = \int_0^1 r_k(t)^2\, dt = 1$$

for every k, while

$$A(0, 0, 1, \ldots, 1) = 0.$$

Therefore A is not weakly sequentially continuous. □

Verification Notes

No computational verification is needed. The proof reduces to three standard facts: Holder's inequality, positivity of the integral form on real L_p lattices, and weak nullity of the Rademacher sequence in finite $L_p[0,1]$.

The only convention-sensitive point is whether Question 3 intended the endpoint $p_i = \infty$. The packet should be read as a full negative answer for the finite-exponent version used in the surrounding paper.